Stanley Nelavala

Software Engineer: Java, Python, React, AWS
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Huntington, IN | Open to relocate anywhere

Summary

Software engineer who builds money-path services: a public Java and Spring Boot payments ledger with idempotency, an outbox and SLOs, a React ops console, a Java QA suite and a C# card dispute service. At work: Python APIs, React apps and GitHub Actions pipelines. Taught Programming II (Java). M.S. Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine that places rooms on a 0.5 ft grid with doors, exits and flow lines; 220+ offline tests on the API.
- Set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission, and raised mobile statement coverage from 14% to 93% (582 tests).

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API, Prometheus, Grafana, k6* GitHub

- Double-entry ledger (authorize, capture, void, refund, transfer) with idempotency keys and a transactional outbox to a Kafka API (Redpanda); property tests and a k6 check that no money is created or lost.
- Three user-facing SLIs with 28-day error budgets and six multi-window burn-rate alerts tested with promtool; a scripted failure lab (Kafka down, Postgres restart, relay crash, poison event, latency injection) with balance checks after each.

Ledger Ops Console and QA Suite | *React, TypeScript, Playwright, axe, Cucumber, Selenium, JMeter* GitHub

- React 19 and TypeScript ops console over read-only endpoints that share one keyset paging contract: 32 Vitest and 13 Playwright tests, 6 axe tests with zero WCAG 2.1 A and AA violations.
- Separate Java QA suite: 41 black-box API tests (16 Cucumber scenarios, REST Assured and JSON-schema contract tests), 6 Selenium UI tests, 0 flaky; JMeter gate passed with about 32,000 samples, 0% errors, p95 81 to 148 ms against 300 ms (local laptop numbers).

Card Dispute Service | *C#, .NET 10, ASP.NET Core, EF Core, PostgreSQL, xUnit, CloudFormation* GitHub

- Dispute state machine where concurrent decisions on one case produce exactly one winner (PostgreSQL xmin optimistic concurrency); idempotent create and keyset paging tested with ties, inserts and deletes; tests written first (30 unit, 32 integration).
- Non-root container image with a clean Trivy scan, Prometheus metrics and readiness checks; CloudFormation for ECS Fargate linted and policy-checked only, with no deploy run.

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C#
Backend: Spring Boot, Flyway, JUnit 5, Testcontainers, FastAPI, ASP.NET Core, EF Core, REST, PostgreSQL, Kafka API
Frontend: React, TypeScript, Sass, Vite, accessibility (axe, WCAG 2.1 A and AA)
Cloud and CI/CD: AWS (ECS, RDS, S3, SQS, GuardDuty), CloudFormation (lint), Terraform, Docker, GitHub Actions
Testing and observability: Cucumber, REST Assured, Selenium, JMeter, Playwright, xUnit, pytest, Prometheus, Grafana, SLOs

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)